Seungyeop Lee

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RESEARCH INTERESTS

Lower-limb wearable robotics Human-robot interaction Vision-based perception

EDUCATION

Korea University — Seoul, Republic of Korea
M.S. in Mechanical Engineering. Integrated Major in Intelligent Mobility. — Sep. 2022 – Feb. 2025
Research Lab: Human-Machine Systems Lab (Prof. Shinsuk Park)
Thesis: Development of an Intuitive Controller and a Monocular Vision-Based Interface for a Disaster-Response UAV

Korea University — Seoul, Republic of Korea
B.S. in Mechanical Engineering. — Mar. 2016 – Aug. 2022
(Note: Absence during Oct. 2017 – Jun. 2019 due to military service)

RESEARCH EXPERIENCE

Samsung Electronics, Robot Vacuum Cleaner SW Lab — Suwon, South Korea
Software Engineer, Spatial AI Research — May 2026 – Present

Spatial AI for Personalized Robot Vacuum Cleaners
- Developed DART, a dual-process architecture that reuses task knowledge and incrementally enriches scene graphs to reduce repeated reasoning.
- Developed observation-viewpoint selection based on object visibility and navigation accessibility.
- Developed a room classifier combining object occurrence likelihoods with visual evidence accumulated during navigation.
- Integrated vision-language model (VLM) reasoning, perception, and navigation in NVIDIA Isaac Sim; simulation evaluations showed 56.2% fewer VLM calls and higher task success (58% vs. 55%) than DART-Frozen.

Iterative Controller Refinement with Simulation Feedback
- Building an LLM-based refinement loop that analyzes simulation feedback and iteratively revises controller code to improve threshold-crossing reliability.
- Evaluating a robot vacuum's threshold-climbing mechanism in NVIDIA Isaac Sim to characterize traversal performance and failure modes.

Wearable Robotics Laboratory — Seoul National University, Seoul, South Korea
Research Intern, Advisor: Prof. Jinsoo Kim — Dec. 2024 – Feb. 2025

Embedded Controller and Sensor Integration for a Soft Exosuit
- Developed an onboard controller for a tendon-driven soft exosuit, implementing a 10-ms sensor-to-motor control loop using multithreaded CAN communication.
- Designed a sensor integration module linking wearable sensors, actuators, and the onboard controller.

Human-Machine Systems Lab — Korea University, Seoul, South Korea
Graduate Researcher, Advisor: Prof. Shinsuk Park — Sep. 2022 – Dec. 2024

Monocular Vision-Based Assistance Interface for Indoor Drones
- Developed a monocular vision-based pilot assistance interface to reduce human error during indoor UAV operation.
- Applied monocular depth estimation for obstacle perception using a single RGB camera on a lightweight UAV.
- Evaluated the interface in a six-participant simulation study: NASA-TLX Mental Demand decreased by approximately 43%, and mean System Usability Scale scores increased from 62.5 to 80 versus an RGB-only interface.

Intelligent Machine Perception and Learning (IMAPLE) Lab — Drexel University, Philadelphia, PA, USA
Visiting Researcher, Advisor: Prof. David Han — Sep. 2023 – Feb. 2024

Sim-to-Real Monocular Depth Estimation
- Applied CycleGAN to translate synthetic training images into the real-image domain for monocular depth estimation.
- Evaluated three depth models on NYU-Depth V2; translated training images reduced absolute relative error by 1.6–3.7% versus unmodified synthetic data.

Autonomous Lighter-Than-Air Robot
- Developed an autonomous aerial robot using STM32 PID control and YOLOv5-based perception.

INDUSTRY EXPERIENCE

Samsung Electronics, Robot Vacuum Cleaner SW Lab — Suwon, South Korea
Software Engineer, Robot Software Development — May 2026 – Present

Hardware-in-the-Loop Simulation and Validation
- Built an Isaac Sim-based HIL test system with Samsung R&D Institute-Delhi, integrating a physical controller board and production control software with simulated robot sensors.

Samsung Electronics, Robot Vacuum Cleaner Development Lab — Suwon, South Korea
Mechanical Engineer, Robot Product Development — Feb. 2025 – Apr. 2026

Robot Vacuum Cleaner Mechanism Development and Mass Production
- Designed and validated mop actuation mechanisms for robot vacuum cleaners, including durability improvement.
- Supported mass production and quality validation, including performance testing and vision-based inspection.

PUBLICATIONS & PRESENTATIONS

** Equal contribution.*

Seungyeop Lee*, Yeeun Kim*, Sooho Park, Seong Oh Lee, and Jong Jin Park. **"DART: From Attentive Reasoning to Intuitive Execution for Experience-Driven Robot Intelligence."** Submitted to the *2027 IEEE International Conference on Robotics and Automation (ICRA). Under review.*

Knut Peterson*, **Seungyeop Lee***, Solmaz Arezoomandan, and David Han. **"Domain-Transferred Synthetic Data Generation for Improving Monocular Depth Estimation"**, *2025 25th International Conference on Control, Automation and Systems (ICCAS)*, 2025.

Seungyeop Lee and Shinsuk Park. **"Obstacle Avoidance for Small Unmanned Aerial Vehicles Using Deep Learning-Based Monocular Depth Estimation"**, *Institute of Control, Robotics and Systems Conference (ICROS)*, *2024. Poster presentation.*

Seungyeop Lee and Shinsuk Park. **"Development of a ROS-Based Indoor Autonomous Quadcopter Platform Using Visual Odometry"**, *Institute of Control, Robotics and Systems Conference (ICROS), 2023. Poster presentation.*

PATENT APPLICATIONS

Seungyeop Lee, Woo Jin Na, Jungil Lee, and Rowoon Lee. **"Station and Cleaning Device for Robot Cleaner."** Korean Patent Application No. 10-2026-0128940, filed Jul. 2026. *Pending.*

Seungyeop Lee, Ahreum Han, Byoungin Lee, and Nguyen Duc Anh. **"Robot Vacuum Cleaner Mop Module Lifting Mechanism."** Korean Patent Application No. 10-2026-0093638, filed May 2026. *Pending.*

Byoungin Lee, Ahreum Han, and **Seungyeop Lee**. **"Cleaner."** Korean Patent Application No. 10-2025-0215312, filed Dec. 2025. *Pending.*

Byoungin Lee, Sangkun Park, **Seungyeop Lee**, and Ahreum Han. **"Robot Cleaner and Method for Controlling Robot Cleaner."** Korean Patent Application No. 10-2025-0166057, filed Nov. 2025. *Pending.*

SELECTED PROJECTS & LEADERSHIP

Horang-Nabi Model Aircraft Club Foundation, *Founder & President* — Korea University, South Korea
Founded Korea University's first model aircraft club and led team building and UAV projects. — Aug. 2019 – Dec. 2021

Korea Robot Aircraft Competition, *Computer Vision Development* — Korea Aerospace Industries Association, South Korea
Developed a pedestrian-tracking UAV integrating onboard perception, target tracking, and flight control. — Apr. 2021 – Sep. 2021

National University Student Model Aircraft Competition, *Team Leader* — Gyeongsang National University, South Korea
Led a team to design and build a scale aircraft for the 21st national university model aircraft competition. — Nov. 2019 – Nov. 2020

Social Economy Engineering Ideas Competition, *Mechanical Design* — University of Seoul, South Korea
Proposed a foldable stair-climbing mobility aid to improve subway accessibility for older adults. — Sep. 2020 – Oct. 2020

Pi village Start-up Team – DRONEDU, *Founder* — Korea University, South Korea
Developed a campus drone education program for high school and university students. — Jan. 2020 – Oct. 2020

AWARDS AND HONORS

Intelligent Robotics Overseas Research Fellowship — Sep. 2023 – Feb. 2024
Received US$20,000 in funding from South Korea's Ministry of Trade, Industry and Energy (MOTIE).

Future Automobile Industry-University Assignment Project Competition — Jan. 2023
Excellence Award, Inter-university Alliance for Future Vehicular Technology (I4FT), South Korea

National University Student Model Aircraft Competition — Nov. 2020
Third Prize, Gyeongsang National University, South Korea

Social Economy Engineering Ideas Competition — Oct. 2020
Third Prize, University of Seoul, South Korea

Creative Comprehensive Design Competition — Sep. 2020
Third Prize, Innovation Center for Engineering Education, Korea University, South Korea

22nd Pi village Start-up Team - DRONEDU — Jun. 2020
First Prize, Korea University, South Korea

SKILLS AND TECHNIQUES

Robotics & Simulation: Python, ROS 1/2, NVIDIA Isaac Sim, Gazebo

Embedded Platforms: Jetson, Raspberry Pi, Arduino

CAD & Prototyping: NX, Fusion 360; laser cutting, 3D printing

TEACHING AND MENTORING

Teaching Assistant, Dynamics – Korea University — Sep. 2024 – Nov. 2024
- Taught two-link manipulator modeling using Adams simulation.

Teaching Assistant, **Robotics – Korea University** — Mar. 2023 – Jun. 2023; Mar. 2024 – Jun. 2024
- Taught Adams simulation, Raspberry Pi, and PID-based DC motor control.

Robotics Career Mentor, Sector Council (SC) Mentoring Program — Mar. 2026
- Led a session in a government-supported career mentoring event as a Samsung Electronics engineer.

Model Aircraft Team Mentor – Korea University — Jan. 2022 – Aug. 2022
- Mentored a model aircraft club team on the structural design of a model aircraft.